

NAMAN SHARMA

Robotics & Computer Vision Engineer | ROS 2 | Edge AI | Simulation

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PROFILE

Final-year Mechatronics Engineer with **12 months of hands-on production deployment** at a Tier-1 automotive supplier. Deep practical fluency in **ROS 2, computer vision, and edge AI** with real robots and real deadlines. Bringing a bias for clean architecture and reliable real-world performance.

EXPERIENCE

Process Engineering Intern — Autoliv India

Feb 2025 – Feb 2026

Tier-1 Automotive Supplier · ISO-Certified · Chandigarh

- Designed and deployed a vision-based OCR inspection system on NVIDIA Jetson Orin Nano Super — ~95% accuracy at 150–250 ms latency per frame under live factory conditions.
- Engineered an industrial thread-detection pipeline (color-based ROI classification); reduced manual inspection workload by ~40%.
- Optimised real-time inference pipelines for sustained throughput within strict edge power and thermal budgets — no GPU server, no cloud dependency.
- Delivered an operator-facing OK/NG GUI with decision logging; adopted by QC team with zero retraining required.
- Collaborated across mechanical, quality, and process teams in an ISO-compliant environment; translated engineering output into auditable documentation.

PROJECTS

Autonomous Mobile Robot — ROS 2 Navigation Stack

ROS 2 · SLAM · D* Lite · Hybrid A*

- Full autonomous navigation stack: ROS 2 Humble, SLAM-based mapping, sensor fusion (LiDAR + encoders), dual-layer path planning with D* Lite global and Hybrid A* local planner.
- Tuned costmaps, obstacle inflation, and recovery behaviours — validated in cluttered indoor environments. Simulated end-to-end in Gazebo before hardware deployment.

Vision-Guided Pick & Place — ROS 2 + Dobot Magician

ROS 2 · OpenCV · Python

- Decoupled vision and control via separate ROS 2 nodes; real-time object localisation and pose estimation using OpenCV.
- Achieved ± 2 mm positional repeatability; eliminated servo jitter through PID gain tuning. Clean pub/sub architecture separating perception, planning, and execution.

TECHNICAL SKILLS

Robotics: ROS 2 (Humble) · SLAM · Sensor Fusion · Motion Planning · URDF · Robot Kinematics · LiDAR · Encoders

Simulation: Gazebo (Fortress / Classic) · Isaac Sim · RViz · URDF/Xacro

Computer Vision: OpenCV · PaddleOCR · ROI Pipelines · Object Detection · Image Preprocessing · OK/NG Logic

Edge AI & Deploy: NVIDIA Jetson Orin Nano Super · TensorRT · ONNX · Real-time Inference Optimisation

Languages: Python (primary) · C++ (working) · Bash

Hardware: Jetson Orin Nano · Raspberry Pi · Arduino · Dobot Magician · Servo/Motor Control

Tools: Git · Linux (Ubuntu) · ISO-compliant documentation · Deterministic system design

EDUCATION

B.E. Mechatronics Engineering — Chitkara University, Punjab

2022 – 2026

Relevant coursework: Robotics · Control Systems · Embedded Systems · Machine Vision · Mechatronics System Design

Portfolio: naman2328.github.io/naman-robotics-vision-portfolio · All projects include live demos or videos